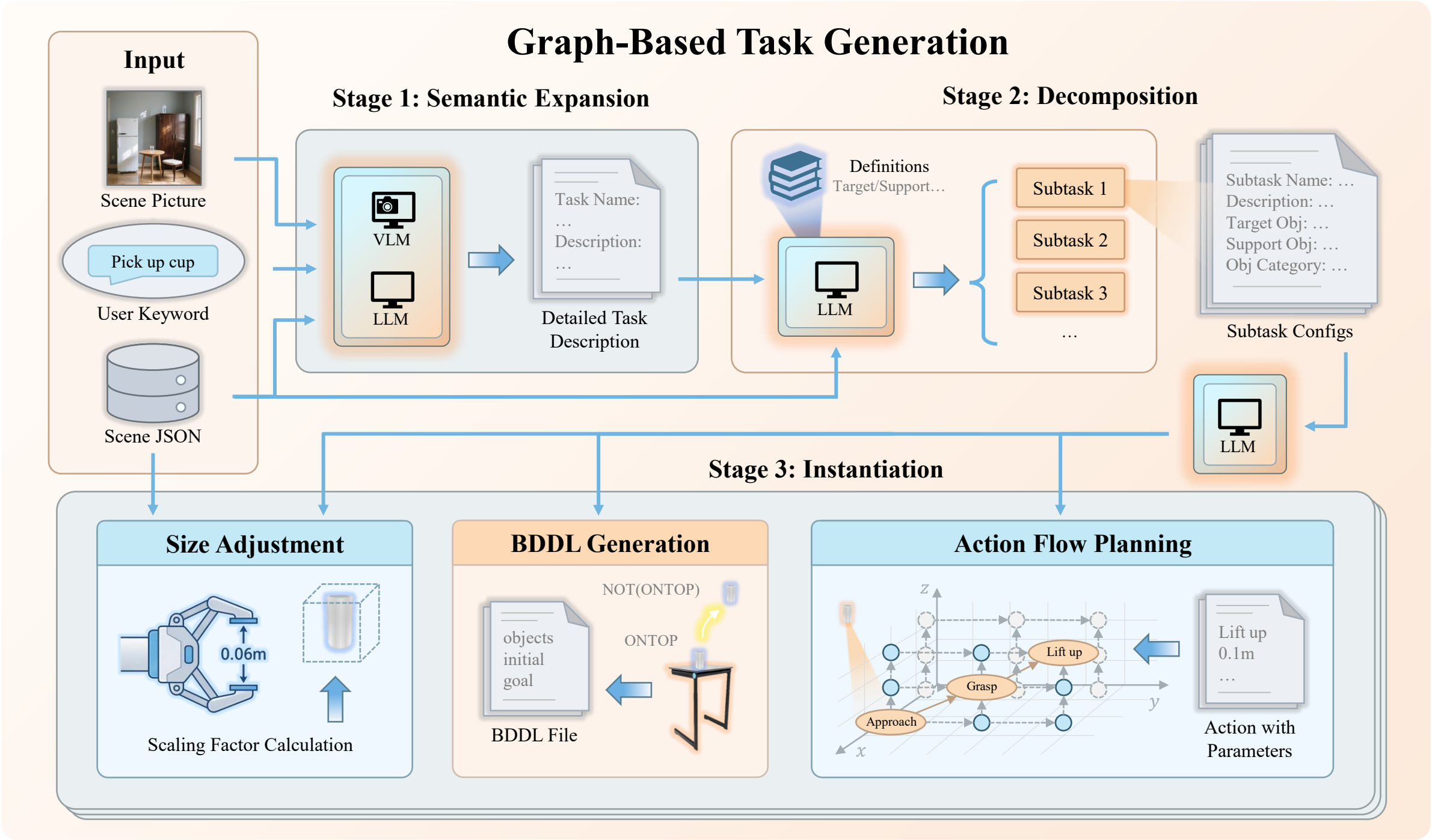
Graph-Based Task Generation
Input
Scene Picture
Pick up cup
User Keyword
Scene JSON
Stage 1: Semantic Expansion
VLM
LLM
Task Name: …
Description: …
Detailed Task Description
Stage 2: Decomposition
Definitions
Target/Support…
LLM
Subtask 1
Subtask 2
Subtask 3
…
Subtask Name: …
Description: …
Target Obj: …
Support Obj: …
Obj Category: …
Subtask Configs
LLM
Stage 3: Instantiation
Size Adjustment
0.06m
Scaling Factor Calculation
BDDL Generation
NOT(ONTOP)
ONTOP
objects
initial
goal
BDDL File
Action Flow Planning
z
y
x
Lift up
Grasp
Approach
Lift up
0.1m
…
Action with Parameters